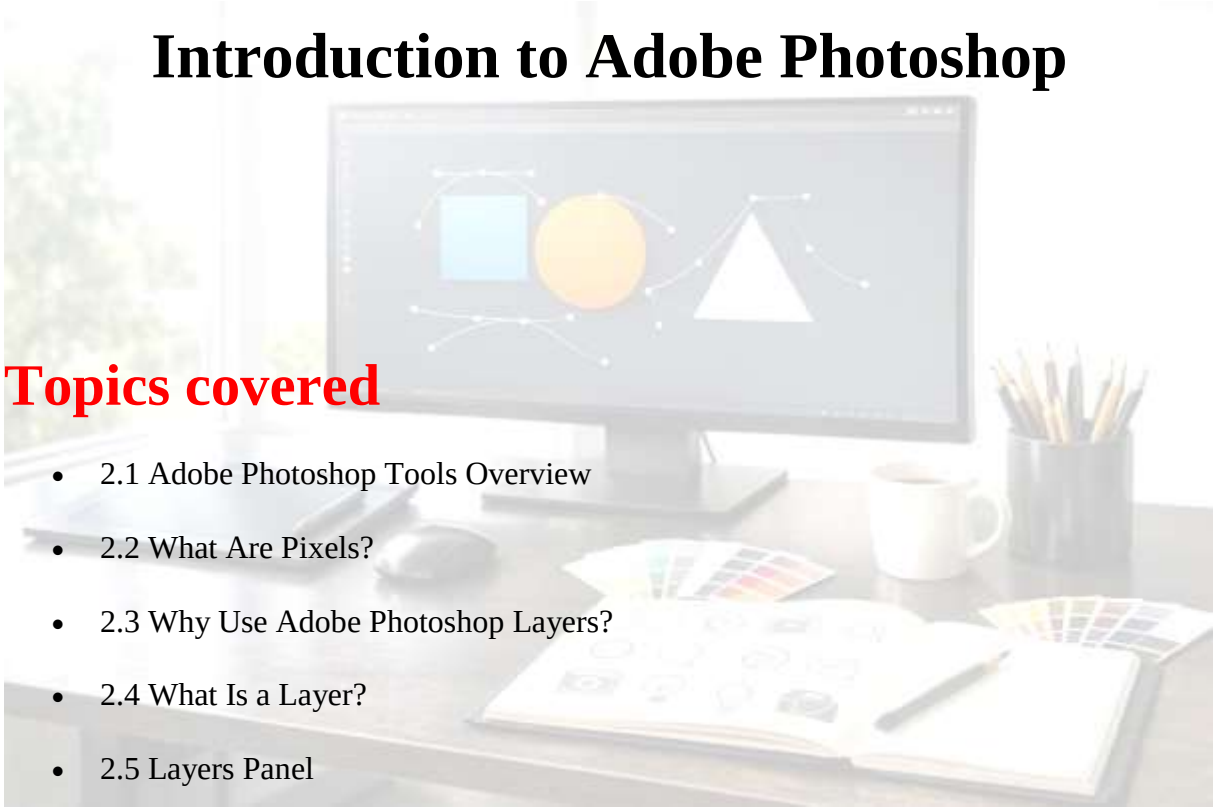
Course Title: Graphic Designing

Course Code: CIT - 344

LECTURE 02:

Introduction to Adobe Photoshop



Topics covered

- 2.1 Adobe Photoshop Tools Overview
- 2.2 What Are Pixels?
- 2.3 Why Use Adobe Photoshop Layers?
- 2.4 What Is a Layer?
- 2.5 Layers Panel
- 2.6 Types of Layers
- 2.7 Layers Shortcut Keys

Prepared by: **Engr. Miss. Ayesha Zulfiqar**

Introduction to Adobe Photoshop

Definition

Adobe Photoshop is a graphics editing software developed by Adobe. It is used to create, edit, enhance, and manipulate digital images and graphic designs.

Explanation

Adobe Photoshop is widely used by graphic designers, photographers, and digital artists. It provides tools for editing photographs, removing backgrounds, adjusting colors, adding text, creating posters, and designing social media posts.

Photoshop allows users to work with different layers, making it easier to edit individual elements without changing the entire design.

Uses of Adobe Photoshop

1. Photo editing and retouching.
2. Background removal and replacement.
3. Poster and banner designing.
4. Social media post designing.
5. Text and typography effects.
6. Digital artwork and image manipulation.
7. Web graphics and advertisements.

Example: A graphic designer can use Photoshop to remove the background of a photograph, add a new background, insert text, and create a promotional poster.

2.1 Adobe Photoshop Tools Overview

Definition

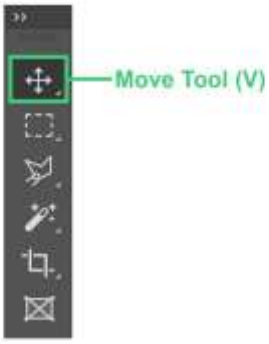
Adobe Photoshop tools are the features used to select, edit, draw, paint, erase, and modify images and graphic elements.

Important Photoshop Tools

1. Move Tool (V)

Used to move selected layers, text, and objects from one position to another.

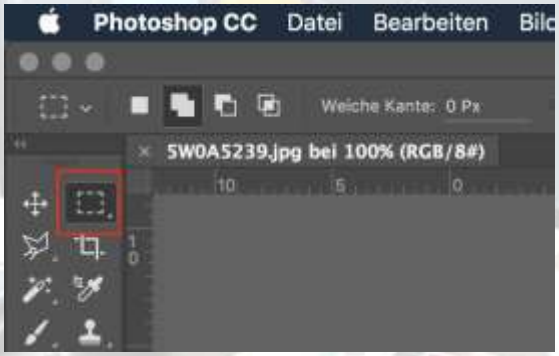
Example: Moving a logo to the top of a poster.



2. Marquee Tool (M)

Used to make selections in rectangular, elliptical, and other supported shapes.

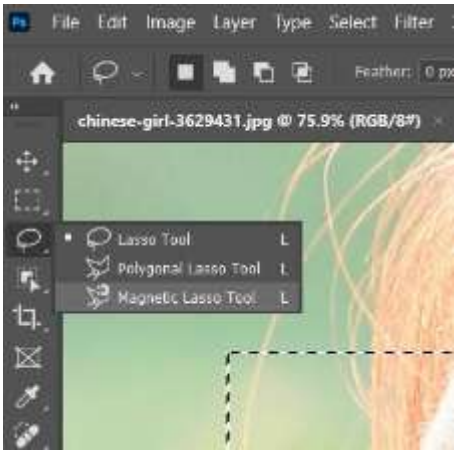
Example: Selecting a rectangular area of a photograph.



3. Lasso Tool (L)

Used to select irregular areas of an image by drawing around them.

Example: Selecting an object with an uneven outline.



4. Crop Tool (C)

Used to remove unwanted outer areas of an image and improve its composition.

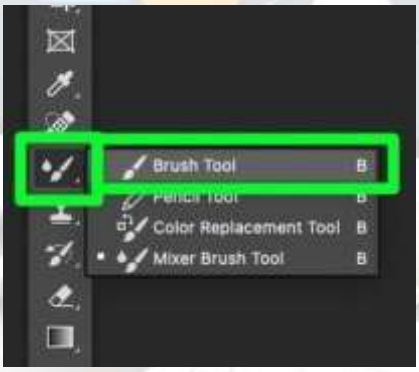
Example: Cropping a photograph to fit a poster.



5. Brush Tool (B)

Used to paint with colors and create brush strokes.

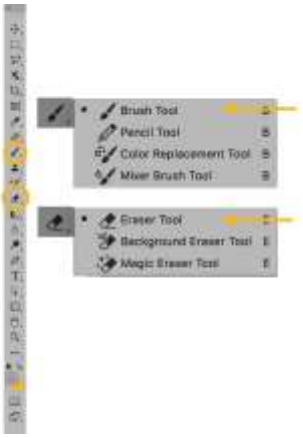
Example: Painting a highlight or adding a creative effect.



6. Eraser Tool (E)

Used to erase pixels from a layer. On a normal pixel layer, erasing generally makes the erased area transparent.

Example: Removing unwanted pixels from an image.



7. Type Tool (T)

Used to add and edit text in a design.

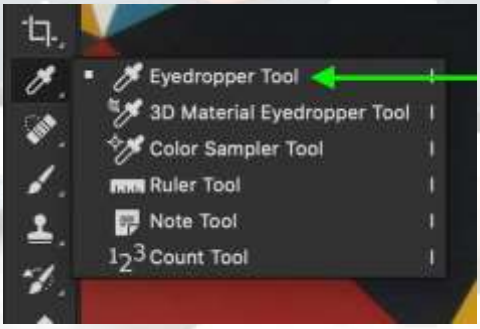
Example: Writing a heading such as “Graphic Design Workshop.”



8. Eyedropper Tool (I)

Used to sample a color from an image.

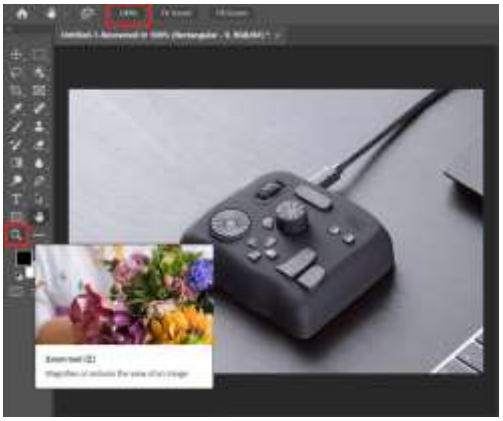
Example: Picking the exact blue color from a logo.



9. Zoom Tool (Z)

Used to zoom in or out to inspect different parts of an image.

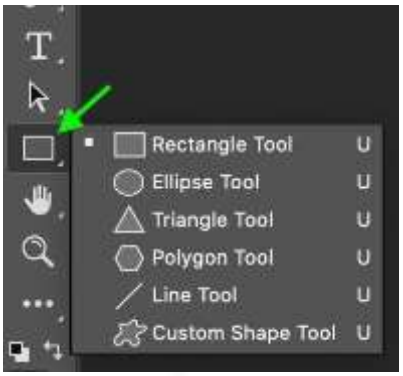
Example: Zooming in to edit small details.



10. Shape Tool (U)

Used to create shapes such as rectangles, circles, lines, and other supported shapes.

Example: Creating a colored box behind poster text.



2.2 What Are Pixels?

Definition

A pixel is the smallest individual picture element in a digital raster image. An image is made up of many tiny pixels arranged in rows and columns.

Explanation

The word pixel comes from “picture element.” Each pixel stores color information that contributes to the complete image. When many pixels are viewed together, they form a photograph or digital graphic.

When an image is enlarged too much, its individual pixels may become visible, making the image look blurry or blocky. This is called pixelation.

Example:

A digital image with dimensions of 100 × 100 pixels contains 10,000 pixels.

\[100 \times 100 = 10,000 \text{ pixels} \]

Important Points

- Pixel means picture element.
- Pixels form raster images.
- Each pixel contains color information.
- More pixels can provide more image detail when the image is captured or created appropriately.
- Enlarging a low-resolution image too much can cause pixelation.

What Is Image Resolution?

Image resolution refers to the amount of image detail, commonly described by pixel dimensions or pixel density, depending on the context.

For example, an image measuring 1920 × 1080 pixels has 1920 pixels horizontally and 1080 pixels vertically.

Remember: Pixels are the tiny elements that make up raster images in Photoshop.

2.3 Why Use Adobe Photoshop Layers?

Definition

Layers are used in Adobe Photoshop to organize different elements of a design separately, allowing each element to be edited independently.

Explanation

Layers make graphic designing easier because text, images, shapes, and other elements can be placed on separate levels. A designer can edit, move, hide, or delete one element without directly changing the other elements.

Layers also allow designers to create complex designs in an organized way.

Advantages of Using Layers

1. Easy editing: Each element can be edited separately.
2. Non-destructive workflow: Adjustments and masks can preserve the original image information.
3. Better organization: Different design elements can be arranged separately.
4. Visibility control: Layers can be hidden or displayed when needed.
5. Flexible design: Elements can be moved, resized, and rearranged.
6. Professional workflow: Complex designs become easier to manage.

Example

Suppose a student creates a college event poster with three layers:

- Layer 1: Background image
- Layer 2: College logo
- Layer 3: Event title

If the student wants to change the event title, the text layer can be edited without replacing the background image.

Remember: Layers make editing easier, flexible, and organized.

2.4 What Is a Layer?

Definition

A layer is an individual level in a Photoshop document that contains image pixels, text, shapes, adjustments, or other design elements.

Explanation

Layers work like transparent sheets placed on top of one another. Each sheet can contain a different part of a design. When these layers are combined, they create the complete image.

The order of layers affects how objects appear. A layer placed above another layer may cover part or all of the content beneath it.

Example

A poster may contain:

- Background layer
- Photograph layer
- Shape layer
- Text layer
- Logo layer

Each layer contains a separate element of the poster.



Important Points

- A layer holds an individual part of a design.
- Layers can be arranged in a particular order.
- Layers can be moved, hidden, and edited.
- A Photoshop document can contain multiple layers.
- Some layers contain pixels, while others contain text, shapes, or adjustments.

Real-life example: Think of layers as transparent pages stacked together. Each page contains one part of a complete drawing.

2.5 Layers Panel

Definition

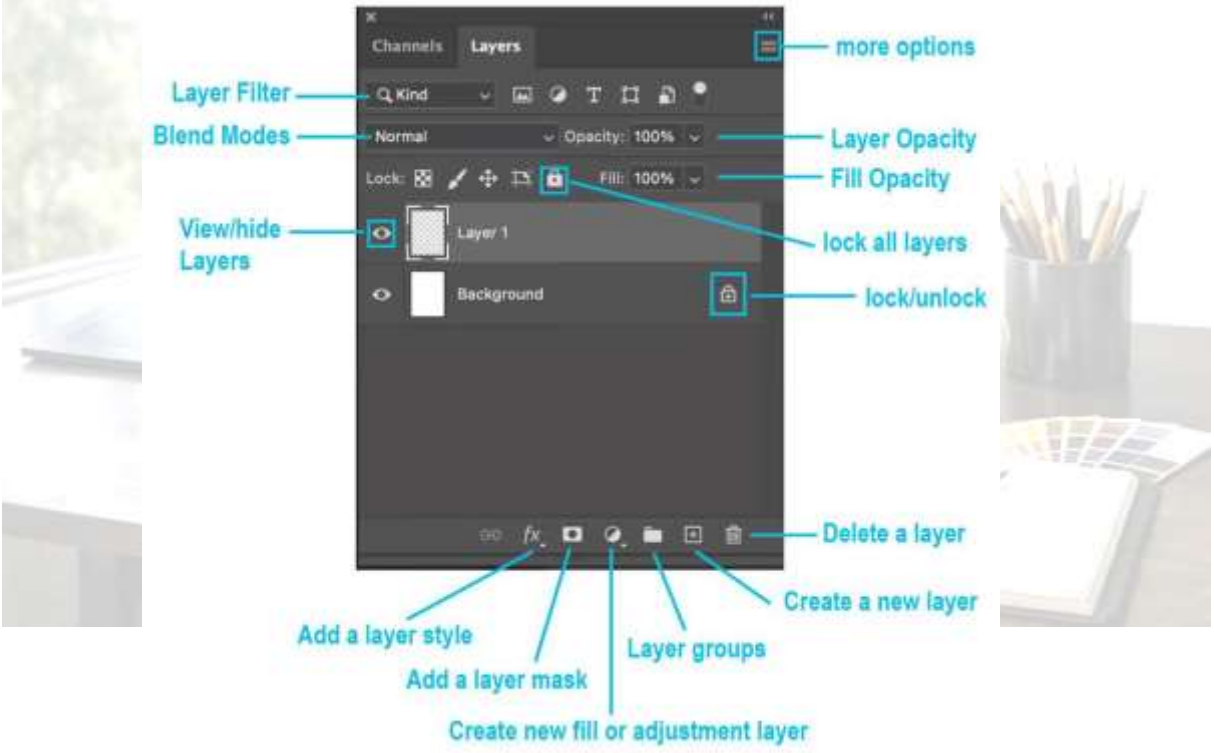
The Layers Panel is a Photoshop workspace panel used to view, create, select, organize, and manage layers in a document.

Explanation

The Layers Panel helps designers control the different elements of an image. It displays the layers in a list and allows users to change their order, visibility, names, and other properties.

The Layers Panel is usually located on the right side of the Photoshop workspace, although its position can be changed.

Main Parts of the Layers Panel



1. Visibility icon (eye): Shows or hides a layer.
2. Layer thumbnail: Displays a small preview of the layer's content.
3. Layer name: Identifies the layer.
4. Selected layer: Indicates which layer is currently active for editing.
5. Opacity: Controls the transparency of a layer.
6. Blend Mode: Controls how a layer interacts visually with the layers below it.
7. Lock options: Restrict certain types of editing on a layer.
8. Add Layer button: Creates a new layer.
9. Delete Layer button: Removes the selected layer.
10. Add Layer Mask button: Adds a mask to control which parts of a layer are visible.

How to Open the Layers Panel

1. Open Adobe Photoshop.
2. Open an existing image or create a new document.
3. Click the Window menu.
4. Select Layers.
5. The Layers Panel will appear.

Shortcut: Press `F7` to show or hide the Layers Panel in standard Photoshop desktop setups.

Uses of the Layers Panel

- Creating new layers.
- Selecting and renaming layers.
- Changing layer order.
- Hiding and showing layers.
- Adjusting opacity.
- Locking and deleting layers.
- Applying masks and managing adjustments.

Remember: The Layers Panel is used to manage all the layers in a Photoshop document.

2.6 Types of Layers in Adobe Photoshop

Photoshop supports different types of layers for different design purposes.

1. Background Layer

A Background Layer is commonly used for the base of a design. It may contain a solid color or an image.

Example: A white background behind a poster design.

Important point: A traditional Photoshop Background Layer has special restrictions, such as limited transparency and movement until it is converted into a normal layer.

2. Normal or Pixel Layer

A pixel layer contains raster image data, such as photographs, painted areas, and other pixels.

Example: A photograph of a person placed in a poster.

Use: Editing, painting, erasing, and manipulating image pixels.

3. Text or Type Layer

A Text Layer contains editable text created with the Type Tool.

Example: The heading “Graphic Designing Workshop.”

Use: Adding titles, headings, captions, and other text.

Important point: Text usually remains editable until it is rasterized or converted into another form.

4. Shape Layer

A Shape Layer contains vector-based shapes created using Photoshop's shape tools.

Example: A circle, rectangle, or line used in a poster.

Use: Creating logos, buttons, borders, icons, and geometric designs.

5. Adjustment Layer

An Adjustment Layer changes the appearance of the layers below it without directly changing their original pixels.

Example: Increasing the brightness or changing the colors of a photograph.

Use: Color correction, brightness, contrast, and other image adjustments.

6. Smart Object Layer

A Smart Object Layer preserves the original content so that transformations and certain filters can be applied more flexibly.

Example: Resizing a logo several times while retaining its original source information.

Use: Flexible editing, reusable graphics, and non-destructive transformations.

7. Layer Group

A Layer Group organizes multiple layers inside one folder in the Layers Panel.

Example: Keeping all the text layers of a poster inside a folder named “Text.”

Use: Organizing complex designs and making layer management easier.

8. Video Layer

A Video Layer is used in supported Photoshop workflows to work with video clips and video-related content.

Example: Editing a video clip in a Photoshop video timeline.

Use: Basic video-related editing and motion graphics workflows.

Quick Revision Table

Layer type	Main purpose
Background Layer	Provides the base of a design
Pixel Layer	Contains raster image pixels
Text Layer	Contains editable text
Shape Layer	Contains vector shapes
Adjustment Layer	Changes visual appearance
Smart Object Layer	Preserves source content for flexible editing
Layer Group	Organizes multiple layers
Video Layer	Works with video content

2.7 Layers Shortcut Keys

Definition

Shortcut keys are keyboard combinations used to perform commands quickly without repeatedly selecting options from menus.

Important Photoshop Layer Shortcuts

The following are common shortcuts for Adobe Photoshop on Windows desktop. Some shortcuts can vary by version, keyboard layout, and workspace configuration.

Shortcut key	Function
`F7`	Show or hide the Layers Panel
`Ctrl + Shift + N`	Create a new layer and open the New Layer dialog
`Ctrl + J`	Duplicate the selected layer or copy selected pixels to a new layer
`Ctrl + G`	Group selected layers
`Ctrl + Shift + G`	Ungroup selected layers
`Ctrl + E`	Merge selected layers

Ctrl + Shift + E	Merge visible layers
Ctrl +]	Move a layer up one level in the stacking order
Ctrl + [	Move a layer down one level in the stacking order
Ctrl + Shift +]	Move a layer to the top
Ctrl + Shift + [	Move a layer to the bottom
Ctrl + T	Enter Free Transform for the selected layer or content
Delete	Delete selected content or a selected layer, depending on the current selection and focus
Ctrl + S	Save the document

Important note about shortcuts

- `Ctrl + J` is particularly useful when duplicating a layer.
- `Ctrl + Shift + N` is useful when creating a new layer.
- `F7` is useful when the Layers Panel is not visible.
- To avoid losing work, save the Photoshop document regularly.

Short questions

1. Define Adobe Photoshop.
2. Write any five uses of Adobe Photoshop.
3. What are Photoshop tools?
4. What is a pixel?
5. What is image resolution?
6. Why do graphic designers use layers?
7. Define a layer in Photoshop.
8. What is the Layers Panel?
9. What is the purpose of the Visibility icon?
10. Define a Text Layer.
11. What is an Adjustment Layer?
12. What is a Smart Object Layer?
13. What is a Layer Group?
14. Write any five Photoshop layer shortcut keys.

Long questions

1. Explain Adobe Photoshop and describe its major uses.
2. Explain any eight tools of Adobe Photoshop with their functions.
3. What are pixels? Explain pixels and image resolution with examples.
4. Define layers and explain the advantages of using layers in Photoshop.
5. Explain the Layers Panel and describe its main parts.
6. Describe the different types of layers in Adobe Photoshop with examples.
7. Write and explain the important shortcut keys used for layer management.

Quick MCQ practice

1. What is the smallest picture element in a raster image?

a) Layer
b) Pixel
c) Folder
d) Font

2. Which shortcut opens or hides the Layers Panel in standard desktop Photoshop?

a) F2
b) F5
c) F7
d) F12

3. Which tool is used to add text?

a) Brush Tool
b) Crop Tool
c) Type Tool
d) Move Tool

4. Which shortcut is commonly used to duplicate a layer?

a) Ctrl + J
b) Ctrl + P
c) Ctrl + Z
d) Ctrl + O

5. Which layer type contains editable text?

a) Background Layer
b) Text Layer
c) Video Layer
d) Group

6. Which layer type helps adjust colors without directly changing the original image pixels?

a) Adjustment Layer
b) Video Layer
c) Background Layer only
d) Text Layer

7. Which panel manages the order and visibility of layers?

a) Color Panel
b) History Panel
c) Layers Panel
d) Character Panel

8. Which tool is commonly used to crop an image?

a) Crop Tool
b) Type Tool
c) Eyedropper Tool
d) Move Tool
e) Reset MCQs